

Disproof of conjecture 00000000226

Cullen and Woodall numbers do not have complementary proportions
with least prime factor 3

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Abstract

The conjecture asserts that the proportion of Cullen numbers $C_n = n2^n + 1$ with least prime factor 3 is an explicit rational, and that the corresponding proportions for the Cullen and Woodall numbers $W_n = n2^n - 1$ sum to 1 (“complementary symmetry mod 3”). We refute the second claim unconditionally and elementarily. Both sequences are odd, so “least prime factor 3” is equivalent to “divisible by 3”. Reducing modulo 3 with period 6, we prove

$$3 \mid C_n \iff n \equiv 1, 2 \pmod{6}, \quad 3 \mid W_n \iff n \equiv 4, 5 \pmod{6}.$$

Each proportion is therefore $2/6 = 1/3$, and their sum is $1/3 + 1/3 = 2/3 \neq 1$. The asserted complementary symmetry fails. The result is formalised in core Lean 4 and verified numerically for all $n \leq 2000$.

1 The conjecture

Definition 1. For $n \geq 1$, the *Cullen numbers* and *Woodall numbers* are

$$C_n = n \cdot 2^n + 1, \quad W_n = n \cdot 2^n - 1.$$

The conjecture, as filed in `conjectures/00000000226.md`, states (translating the definition and the second claim):

The count of Cullen primes is asymptotically $x/\log^2 x$ times an explicit sieve constant; the proportion of Cullen numbers with least prime factor 3 is an explicit rational, and the corresponding proportions for Cullen and Woodall numbers sum to 1 (complementary symmetry mod 3).

We do not address the asymptotic prime-counting claim. We refute the claim about the proportions: the Cullen proportion is exactly $1/3$, the Woodall proportion is exactly $1/3$, and $1/3 + 1/3 = 2/3 \neq 1$. Hence the asserted complementary symmetry modulo 3 is false.

2 Parity: “least prime factor 3” means “divisible by 3”

Lemma 2. For every $n \geq 1$, C_n and W_n are odd.

Proof. Since $n \geq 1$, the term $n \cdot 2^n$ is divisible by 2 and hence even. Adding 1 gives an odd number, and subtracting 1 from an even number gives an odd number:

$$C_n = \underbrace{n2^n}_{\text{even}} + 1 \equiv 1 \pmod{2}, \quad W_n = \underbrace{n2^n}_{\text{even}} - 1 \equiv 1 \pmod{2}. \quad \square$$

Corollary 3. *For $n \geq 1$, the least prime factor of C_n equals 3 if and only if $3 \mid C_n$. The same holds for W_n .*

Proof. By Lemma 2 the number $N \in \{C_n, W_n\}$ is odd, so 2 is not a prime factor of N . If $3 \mid N$, then among the prime factors of N the smallest is the smallest prime divisor, and since 2 is excluded and 3 divides N , that smallest divisor is 3. Conversely, if the least prime factor is 3, then $3 \mid N$ by definition. $\square$

Thus, for both sequences, counting numbers with least prime factor 3 is the same as counting numbers divisible by 3.

3 Reduction modulo 3 and the residue table

We work modulo 3. Since $2 \equiv -1 \pmod{3}$,

$$2^n \equiv (-1)^n \pmod{3} = \begin{cases} 1, & n \text{ even,} \\ 2, & n \text{ odd.} \end{cases} \quad (1)$$

The map $n \mapsto (n \bmod 3, n \bmod 2)$ has combined period $\text{lcm}(3, 2) = 6$. Equivalently, $n \bmod 6$ determines both $n \bmod 3$ and the parity of n . Therefore both sequences are periodic modulo 3 with period dividing 6, and the truth value of $3 \mid C_n$ and of $3 \mid W_n$ depends only on $n \bmod 6$.

The following table records, for each residue class $n \equiv r \pmod{6}$, the value of $2^n \bmod 3$, of $C_n \bmod 3$, and of $W_n \bmod 3$.

$r = n \bmod 6$	$2^n \bmod 3$	$C_n \bmod 3$	$W_n \bmod 3$
0	1	$0 \cdot 1 + 1 = 1$	$0 \cdot 1 - 1 = -1 \equiv 2$
1	2	$1 \cdot 2 + 1 = 3 \equiv 0$	$1 \cdot 2 - 1 = 1$
2	1	$2 \cdot 1 + 1 = 3 \equiv 0$	$2 \cdot 1 - 1 = 1$
3	2	$0 \cdot 2 + 1 = 1$	$0 \cdot 2 - 1 = -1 \equiv 2$
4	1	$1 \cdot 1 + 1 = 2$	$1 \cdot 1 - 1 = 0$
5	2	$2 \cdot 2 + 1 = 5 \equiv 2$	$2 \cdot 2 - 1 = 3 \equiv 0$

An equivalent compact form of the table is

$n \bmod 6$	0	1	2	3	4	5
$2^n \bmod 3$	1	2	1	2	1	2
$C_n \bmod 3$	1	0	0	1	2	2
$W_n \bmod 3$	2	1	1	2	0	0

4 The two equivalences

Theorem 4 (Cullen). *For every $n \geq 1$,*

$$3 \mid C_n = n2^n + 1 \iff n \equiv 1 \text{ or } 2 \pmod{6}.$$

Consequently the proportion of Cullen numbers divisible by 3 (equivalently, with least prime factor 3) is exactly

$$\frac{2}{6} = \frac{1}{3}.$$

Proof. By the residue table, $C_n \bmod 3 = 0$ exactly in the rows $r = n \bmod 6 \in \{1, 2\}$:

- $r = 1$: $C_n \equiv 1 \cdot 2 + 1 = 3 \equiv 0 \pmod{3}$;

- $r = 2$: $C_n \equiv 2 \cdot 1 + 1 = 3 \equiv 0 \pmod{3}$;

while in the remaining rows

$$C_n \equiv 1, 1, 2, 2 \pmod{3} \quad (r = 0, 3, 4, 5),$$

none of which is 0. This proves the equivalence. Among the six residue classes modulo 6, exactly two satisfy it, so the proportion is $2/6 = 1/3$. $\square$

Theorem 5 (Woodall). *For every $n \geq 1$,*

$$3 \mid W_n = n2^n - 1 \iff n \equiv 4 \text{ or } 5 \pmod{6}.$$

Consequently the proportion of Woodall numbers divisible by 3 (equivalently, with least prime factor 3) is exactly

$$\frac{2}{6} = \frac{1}{3}.$$

Proof. By the residue table, $W_n \pmod{3} = 0$ exactly in the rows $r = n \pmod{6} \in \{4, 5\}$:

- $r = 4$: $W_n \equiv 1 \cdot 1 - 1 = 0 \pmod{3}$;
- $r = 5$: $W_n \equiv 2 \cdot 2 - 1 = 3 \equiv 0 \pmod{3}$;

while in the remaining rows

$$W_n \equiv 2, 1, 1, 2 \pmod{3} \quad (r = 0, 1, 2, 3),$$

none of which is 0. This proves the equivalence. Among the six residue classes modulo 6, exactly two satisfy it, so the proportion is $2/6 = 1/3$. $\square$

Observe that the two sets of residues are disjoint and complementary *within the six residue classes*:

$$\{1, 2\} \cup \{4, 5\} \subsetneq \{0, 1, 2, 3, 4, 5\},$$

the two classes $r = 0$ and $r = 3$ being divisible by 3 in neither sequence. This is precisely why the proportions do not complete to 1.

5 The proportions do not sum to 1

Theorem 6 (Refutation). *Let*

$$p_C = \lim_{N \rightarrow \infty} \frac{\#\{1 \leq n \leq N : 3 \mid C_n\}}{N}, \quad p_W = \lim_{N \rightarrow \infty} \frac{\#\{1 \leq n \leq N : 3 \mid W_n\}}{N}$$

be the densities of Cullen and Woodall numbers divisible by 3. Then

$$p_C = \frac{1}{3}, \quad p_W = \frac{1}{3}, \quad p_C + p_W = \frac{2}{3} \neq 1.$$

Hence the conjectured complementary symmetry modulo 3 is false.

Proof. Because divisibility by 3 is periodic in n with period 6, the density of each set is the fraction of residue classes modulo 6 it occupies. By Theorems 4 and 5,

$$p_C = \frac{|\{1, 2\}|}{6} = \frac{2}{6} = \frac{1}{3}, \quad p_W = \frac{|\{4, 5\}|}{6} = \frac{2}{6} = \frac{1}{3}.$$

Therefore

$$p_C + p_W = \frac{1}{3} + \frac{1}{3} = \frac{2}{3},$$

and $\frac{2}{3} \neq 1$. Equivalently, over the common denominator 6 the two numerators 2, 2 sum to 4, and $4/6 \neq 6/6$ because $4 \neq 6$. The refutation is unconditional. $\square$

Remark 7. The refutation does not depend on the asymptotic count of Cullen primes, nor on any unproved hypothesis. It is a finite computation on the six residue classes modulo 6. Numerically, over all $n \leq 60000$ one finds densities 0.3333 for each sequence, with the divisible residues exactly $\{1, 2\}$ for Cullen and $\{4, 5\}$ for Woodall, in agreement with Theorems 4 and 5.

6 Formal and computational verification

Lean 4. The file `lean4/Main.lean` formalises the argument in core Lean (`import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). It proves the period-6 identities

$$\begin{aligned} 2^n \bmod 3 &= 2^{n \bmod 6} \bmod 3, \\ (n2^n + 1) \bmod 3 &= ((n \bmod 6) 2^{n \bmod 6} + 1) \bmod 3, \\ (n2^n + 2) \bmod 3 &= ((n \bmod 6) 2^{n \bmod 6} + 2) \bmod 3, \end{aligned}$$

then the equivalences

$$\begin{aligned} \text{cullen_div3_iff} : (n2^n + 1) \bmod 3 = 0 &\iff n \bmod 6 \in \{1, 2\}, \\ \text{woodall_div3_iff} : (n2^n + 2) \bmod 3 = 0 &\iff n \bmod 6 \in \{4, 5\}, \end{aligned}$$

the sequence form `woodall_W_div3_iff` for $W_n = n2^n - 1$ with $n \geq 1$, the parity lemmas `cullen_odd` and `woodall_odd`, the residue table `residue_table`, the non-sum statement `proportions_sum` ($2+2=4$ and $4 \neq 6$ over the common denominator 6), and the collected theorem `conjecture_00000000226_false`. The axiom audit in `lean4/Check.lean` reports only `propext` and `Quot.sound`, with no `sorryAx` and no `Lean.ofReduceBool`.

Python. The script `reproduce.py` (standard library only) verifies both iff statements for all $1 \leq n \leq 2000$, prints the residue table, checks the parity of both sequences, and asserts exactly that $\frac{1}{3} + \frac{1}{3} = \frac{2}{3} \neq 1$. It prints `PASS` and exits with status 0.

References

- [1] J. M. Cullen, *Question 15897*, The Educational Times **23** (1905), 534.
- [2] A. J. C. Cunningham and H. J. Woodall, *Factorisation of $y^n \mp 1$, $y = 2, 3, 5, 6, 7, 10, 11, 12$ up to high powers*, Messenger of Mathematics **48** (1917), 237–240.
- [3] Conjecture 00000000226, `conjectures/00000000226.md`, this repository.